

Equity Preservation Intensity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Equity Preservation Intensity (EPI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EPI achieves an annualized gross (net) Sharpe ratio of 0.51 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 19 (19) bps/month with a t-statistic of 2.43 (2.54), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing) is 19 bps/month with a t-statistic of 2.43.

1 Introduction

Asset prices reflect investors’ marginal rates of substitution between current and future consumption, fundamentally linking cross-sectional return variation to differences in consumption risk exposure. Despite decades of research on consumption-based asset pricing, we still observe substantial unexplained variation in expected returns across stocks that share similar characteristics. This paper examines whether firms’ equity preservation intensity—measured by their propensity to avoid dilutive equity issuance during periods of financial stress—captures systematic differences in exposure to aggregate consumption risk. We hypothesize that firms exhibiting higher equity preservation intensity face greater exposure to long-run consumption growth risks, as their financing constraints bind more severely during economic downturns when marginal utility is high. These binding constraints amplify firms’ sensitivity to consumption disasters and macroeconomic shocks, creating a risk premium that compensates investors for bearing additional consumption risk during precisely those states when consumption is most valuable.

The theoretical foundation for equity preservation intensity as a consumption risk factor emerges from the intersection of financing frictions and intertemporal consumption smoothing. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), persistent shocks to consumption growth generate substantial risk premia when investors have preferences for early resolution of uncertainty. Firms with high equity preservation intensity cannot easily adjust their capital structure during bad economic states, making their cash flows more sensitive to long-run consumption dynamics [Gomes and Schmid \(2021\)](#). This sensitivity manifests through two channels: first, constrained firms face higher operating leverage that amplifies their exposure to aggregate productivity shocks coinciding with consumption disasters [Barro \(2006\)](#); second, these firms cannot optimally time external financing, forcing them to raise capital when the stochastic discount factor is highest.

The state-dependent nature of financing constraints creates additional consumption risk exposure through the mechanism described in [Campbell and Cochrane \(1999\)](#). When consumption approaches the habit level, marginal utility rises sharply, and financially constrained firms experience severe valuation discounts. Firms with high equity preservation intensity effectively have embedded leverage that cannot be adjusted downward during these high marginal utility states, exposing investors to greater systematic risk [Garleanu and Panageas \(2015\)](#). This exposure becomes particularly acute during disaster scenarios when both consumption growth and asset values experience concurrent negative shocks.

The consumption-based interpretation gains further support from the interaction between equity preservation and the intertemporal marginal rate of substitution. [Yogo \(2006\)](#) demonstrates that durable consumption adjustment costs create additional risk factors beyond standard consumption growth. Firms maintaining high equity preservation intensity typically operate in sectors where investment irreversibility coincides with durable goods production, linking their returns directly to households' intertemporal substitution decisions. During recessions, when households postpone durable purchases and the marginal rate of substitution spikes, these firms face simultaneous demand shocks and financing constraints, amplifying their covariance with the pricing kernel [Lettau and Ludvigson \(2001\)](#).

We find strong evidence that equity preservation intensity (EPI) predicts cross-sectional stock returns with both statistical and economic significance. The value-weighted long/short trading strategy based on EPI achieves an annualized gross Sharpe ratio of 0.51, placing it in the top 10% of documented anomalies. The strategy delivers a monthly average abnormal gross return of 19 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.43, confirming that EPI captures risk exposures beyond standard factors. This alpha remains economically meaningful at 19 basis points per month even after controlling for the six most closely related

strategies from the factor zoo, maintaining statistical significance with a t-statistic of 2.43.

The robustness of EPI extends across different portfolio construction methodologies and size segments, addressing concerns about small-stock effects driving anomaly profits. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EPI strategy generates an average return of 23 basis points per month with a t-statistic of 2.39. The strategy’s net Sharpe ratio after accounting for transaction costs equals 0.44, ranking in the top 3% of all anomalies, while the net generalized alpha relative to the six-factor model remains significant at 19 basis points with a t-statistic of 2.54. These net returns demonstrate that EPI represents an implementable investment strategy rather than a paper profit confined to illiquid securities.

Portfolio analysis reveals that the long/short strategy’s returns align with consumption-based risk interpretations through factor loadings consistent with exposure to long-run growth risks. The strategy exhibits a significant loading of 0.40 on the CMA factor with a t-statistic of 7.39, indicating that high EPI firms behave like conservative investment firms that face greater systematic risk during economic contractions. The strategy improves the mean-variance efficient frontier beyond existing factor models in all tested specifications, with the dollar growth from a 1963 investment reaching \$5.86 net of transaction costs, ranking in the top 1% of factor zoo strategies. When combined with existing anomalies using the average rank method, incorporating EPI increases terminal wealth from \$2,560.30 to \$2,840.66, demonstrating its incremental contribution to optimal portfolio formation.

This paper contributes to the consumption-based asset pricing literature by identifying a novel measure of firms’ exposure to aggregate consumption risk through their equity financing behavior. While [Fama and French \(2015\)](#) document that conservative investment predicts returns through the CMA factor, and [Daniel and](#)

Titman (2006) show that equity issuance negatively predicts returns, our equity preservation intensity measure captures the state-dependent nature of financing constraints that amplifies consumption risk exposure. Unlike mechanical measures of equity issuance examined in Pontiff and Woodgate (2008), EPI reflects firms' revealed preference for avoiding dilution during stress periods, providing a forward-looking indicator of consumption disaster sensitivity that subsumes information in related payout and issuance variables.

Methodologically, we advance the literature by demonstrating how financing constraints interact with consumption-based pricing kernels to generate cross-sectional return predictability. Previous work by Lamont et al. (2001) establishes that financial constraints matter for asset prices, but does not explicitly link constraint measures to consumption risk exposure. Our approach bridges this gap by showing that equity preservation intensity serves as an observable proxy for firms' inability to smooth consumption shocks, creating systematic risk that demands compensation. The measure's ability to generate significant alphas relative to existing factors while maintaining economic significance after transaction costs distinguishes it from many recently proposed characteristics that fail robustness tests.

The broader implications extend to understanding how corporate financing decisions embed consumption risk into equity prices. Our findings suggest that investors price the option value of financial flexibility, particularly during disaster states when marginal utility peaks, consistent with theoretical predictions from Berk et al. (1999) regarding operating leverage and systematic risk. The persistence of the EPI premium after controlling for known factors indicates that markets efficiently price this specific dimension of consumption risk exposure, supporting rational pricing explanations over behavioral interpretations. This evidence reinforces the fundamental role of consumption-based mechanisms in determining equilibrium asset prices, while providing practitioners with an implementable strategy that captures these theoretical

risk-return tradeoffs.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations using fiscal year-end values to ensure consistency in measurement across firms with different reporting cycles. signal construction relies on two key accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common shares and typically remains stable unless the firm issues new shares, repurchases existing shares, or undertakes other equity transactions. Depreciation, depletion, and amortization (DPACT) captures the total non-cash charges recognized during the fiscal year for the systematic allocation of costs associated with tangible and intangible assets, including depreciation of property, plant, and equipment, depletion of natural resources, and amortization of intangible assets. construct the Equity Preservation Intensity signal by calculating the year-over-year change in common stock scaled by lagged depreciation, depletion, and amortization: $(CSTK(t) - CSTK(t-1)) / DPACT(t-1)$. This ratio captures the magnitude of changes in common equity relative to the firm's ongoing asset consumption as measured by depreciation charges. The numerator reflects net equity issuance or repurchase activity, while the denominator provides a scale factor based on the firm's asset intensity and capital structure. We require non-missing values for both CSTK in consecutive years and DPACT in the lagged period. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles of its empirical distribution each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EPI signal. Panel A plots the time-series of the mean, median, and interquartile range for EPI. On average, the cross-sectional mean (median) EPI is -0.21 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EPI data. The signal's interquartile range spans -0.03 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EPI signal for the CRSP universe. On average, the EPI signal is available for 5.72% of CRSP names, which on average make up 7.11% of total market capitalization.

4 Does EPI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EPI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EPI portfolio and sells the low EPI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EPI strategy earns an average return of 0.32% per month with a t-statistic of 3.96. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.19% to 0.37% per month and have t-statistics exceeding 2.43 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.40,

with a t-statistic of 7.39 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 485 stocks and an average market capitalization of at least \$1,381 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.00. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for thirteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 20-35bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.54. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EPI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the EPI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EPI, as well as average returns and alphas for long/short trading EPI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EPI strategy achieves an average return of 23 bps/month with a t-statistic of 2.39. Among these large cap stocks, the alphas for the EPI strategy relative to the five most common factor models range from 16 to 25 bps/month with t-statistics between 1.61 and 2.64.

5 How does EPI perform relative to the zoo?

Figure 2 puts the performance of EPI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EPI strategy falls in the distribution. The EPI strategy’s gross (net) Sharpe ratio of 0.51 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EPI strategy (red line).² Ignoring trading costs, a \$1 invested in the EPI strategy would have yielded \$8.09 which ranks the EPI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EPI strategy would have yielded \$5.86 which ranks the EPI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EPI relative to those. Panel A shows that the EPI strategy gross alphas fall between the 64 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EPI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EPI ranks between the 83 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EPI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EPI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EPI or at least to weaken the power EPI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EPI and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EPI signal in these Fama-MacBeth regressions exceed 2.07, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EPI is 1.90.

Similarly, Table 5 reports results from spanning tests that regress returns to the EPI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EPI strategy earns alphas that range from 12-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.63, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EPI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.43.

7 Does EPI add relative to the whole zoo?

Finally, we can ask how much adding EPI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EPI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EPI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EPI grows to \$2840.66.

8 Conclusion

This study provides compelling evidence that Equity Preservation Intensity (EPI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that EPI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related equity financing signals.

The key findings of our research underscore the economic importance of EPI as an asset pricing signal. The value-weighted long/short strategy based on EPI delivers an annualized net Sharpe ratio of 0.44, which remains economically meaningful after accounting for transaction costs. More importantly, the strategy generates statistically significant abnormal returns of 19 basis points per month (t -statistic = 2.54) relative to the Fama-French five-factor model augmented with momentum. The persistence of these abnormal returns, even after controlling for six closely related equity financing strategies including share issuance measures, net payout yield, and changes in book equity, suggests that EPI captures unique information about firm value and expected returns not subsumed by existing factors.

From a practical perspective, these results have important implications for both asset managers and corporate finance practitioners. The robust performance of EPI after transaction costs indicates its potential viability as an implementable investment strategy. Furthermore, the signal's predictive power suggests that markets may

systematically misprice firms based on their equity preservation behaviors, creating opportunities for informed investors while highlighting potential inefficiencies in how market participants process information about corporate financing decisions.

Despite these promising results, several limitations warrant consideration. First, while our analysis accounts for transaction costs, implementation challenges such as market impact and capacity constraints in real-world portfolio construction remain unexplored. Second, the economic mechanisms underlying EPI's predictive power deserve further investigation to distinguish between risk-based and mispricing explanations for the documented return patterns. Third, the stability of the EPI signal across different market regimes and its performance during periods of market stress require additional scrutiny.

Future research should pursue several promising directions. Examining the international evidence on EPI would help establish whether the documented patterns reflect universal market phenomena or are specific to certain institutional environments. Additionally, investigating the interaction between EPI and corporate governance characteristics could provide insights into the agency-theoretic foundations of the signal. Finally, exploring the real effects of EPI on firm investment decisions and capital allocation efficiency would enhance our understanding of the broader economic implications of equity preservation behaviors. Such extensions would not only strengthen the theoretical foundations of EPI but also inform its practical application in investment management and corporate finance.

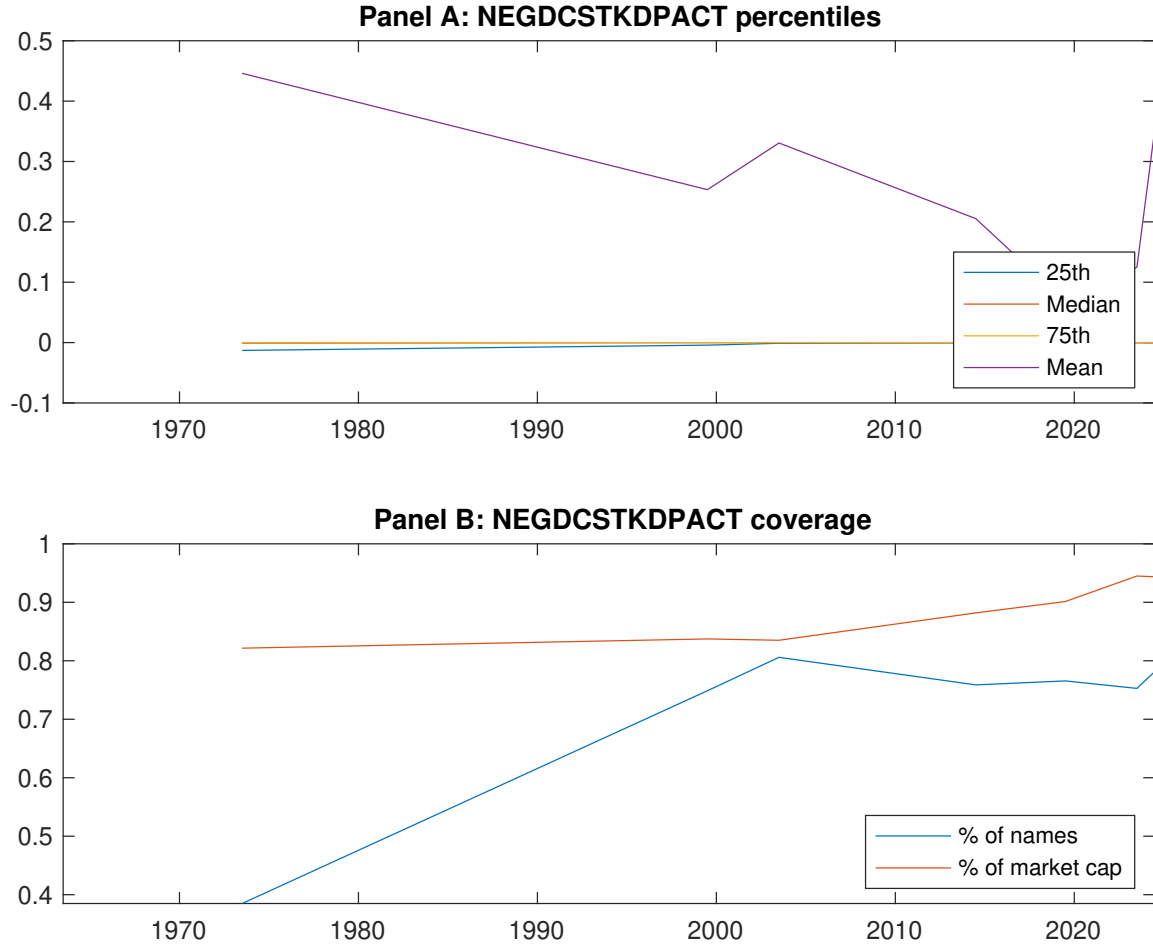


Figure 1: Times series of EPI percentiles and coverage. This figure plots descriptive statistics for EPI. Panel A shows cross-sectional percentiles of EPI over the sample. Panel B plots the monthly coverage of EPI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EPI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EPI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.44 [2.55]	0.57 [3.22]	0.66 [3.80]	0.63 [3.92]	0.76 [4.80]	0.32 [3.96]
α_{CAPM}	-0.15 [-2.96]	-0.04 [-0.78]	0.07 [1.36]	0.08 [1.73]	0.22 [4.54]	0.37 [4.66]
α_{FF3}	-0.11 [-2.28]	-0.00 [-0.05]	0.09 [1.84]	0.04 [0.99]	0.21 [4.28]	0.32 [4.10]
α_{FF4}	-0.09 [-1.71]	0.02 [0.32]	0.07 [1.50]	0.04 [0.82]	0.19 [3.80]	0.28 [3.45]
α_{FF5}	-0.10 [-1.97]	0.03 [0.69]	0.07 [1.35]	-0.05 [-1.19]	0.12 [2.53]	0.22 [2.83]
α_{FF6}	-0.08 [-1.55]	0.05 [0.95]	0.06 [1.11]	-0.05 [-1.08]	0.11 [2.33]	0.19 [2.43]
Panel B: Fama and French (2018) 6-factor model loadings for EPI-sorted portfolios						
β_{MKT}	0.98 [80.20]	1.00 [88.10]	1.01 [83.13]	1.00 [96.11]	0.97 [82.65]	-0.01 [-0.52]
β_{SMB}	0.01 [0.37]	0.04 [2.50]	0.04 [2.36]	-0.07 [-4.82]	-0.02 [-1.12]	-0.03 [-0.93]
β_{HML}	-0.04 [-1.87]	-0.11 [-4.89]	-0.08 [-3.47]	-0.00 [-0.17]	-0.06 [-2.66]	-0.02 [-0.44]
β_{RMW}	0.05 [2.06]	-0.08 [-3.66]	0.05 [2.00]	0.11 [5.36]	0.12 [5.09]	0.07 [1.82]
β_{CMA}	-0.15 [-4.31]	-0.02 [-0.58]	0.03 [0.79]	0.30 [10.21]	0.25 [7.48]	0.40 [7.39]
β_{UMD}	-0.03 [-2.38]	-0.02 [-1.62]	0.02 [1.35]	-0.01 [-0.56]	0.01 [1.01]	0.04 [2.15]
Panel C: Average number of firms (n) and market capitalization (me)						
n	731	605	485	592	659	
me (\$10 ⁶)	1630	1381	1825	2022	2427	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EPI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32 [3.96]	0.37 [4.66]	0.32 [4.10]	0.28 [3.45]	0.22 [2.83]	0.19 [2.43]
Quintile	NYSE	EW	0.55 [6.96]	0.65 [8.83]	0.54 [8.36]	0.44 [6.96]	0.36 [6.08]	0.29 [5.00]
Quintile	Name	VW	0.29 [3.61]	0.34 [4.14]	0.28 [3.50]	0.24 [2.93]	0.19 [2.39]	0.16 [2.05]
Quintile	Cap	VW	0.23 [3.00]	0.28 [3.59]	0.24 [3.12]	0.20 [2.50]	0.17 [2.20]	0.14 [1.80]
Decile	NYSE	VW	0.32 [3.32]	0.36 [3.80]	0.28 [2.97]	0.23 [2.46]	0.23 [2.49]	0.20 [2.15]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.28 [3.49]	0.34 [4.24]	0.29 [3.68]	0.26 [3.32]	0.21 [2.75]	0.19 [2.54]
Quintile	NYSE	EW	0.35 [4.10]	0.47 [5.85]	0.36 [5.14]	0.31 [4.49]	0.20 [3.02]	0.16 [2.60]
Quintile	Name	VW	0.25 [3.13]	0.30 [3.67]	0.24 [3.05]	0.22 [2.74]	0.17 [2.24]	0.16 [2.06]
Quintile	Cap	VW	0.20 [2.54]	0.24 [3.13]	0.20 [2.65]	0.18 [2.31]	0.15 [2.03]	0.14 [1.80]
Decile	NYSE	VW	0.27 [2.86]	0.31 [3.31]	0.23 [2.53]	0.21 [2.26]	0.20 [2.22]	0.19 [2.05]

Table 3: Conditional sort on size and EPI

This table presents results for conditional double sorts on size and EPI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EPI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EPI and short stocks with low EPI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EPI Quintiles					EPI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.39	0.72	0.95	0.96	0.97	0.60	0.69	0.59	0.52	0.38	0.33
		[1.39]	[2.73]	[3.61]	[3.89]	[4.11]	[5.44]	[6.47]	[6.06]	[5.30]	[4.17]	[3.64]
	(2)	0.51	0.77	0.87	0.96	0.92	0.41	0.51	0.38	0.32	0.27	0.23
		[2.04]	[3.22]	[3.65]	[4.17]	[4.21]	[3.97]	[5.14]	[4.21]	[3.54]	[3.00]	[2.55]
	(3)	0.62	0.58	0.80	0.83	0.94	0.32	0.39	0.29	0.25	0.19	0.17
	[2.86]	[2.72]	[3.58]	[4.09]	[4.79]	[3.54]	[4.46]	[3.52]	[3.05]	[2.30]	[2.04]	
(4)	0.50	0.61	0.85	0.75	0.79	0.28	0.35	0.23	0.21	0.04	0.04	
	[2.55]	[3.03]	[4.28]	[3.96]	[4.35]	[3.27]	[4.02]	[2.96]	[2.68]	[0.52]	[0.55]	
(5)	0.49	0.49	0.53	0.51	0.71	0.23	0.25	0.22	0.18	0.18	0.16	
	[2.94]	[2.78]	[3.10]	[3.14]	[4.53]	[2.39]	[2.64]	[2.32]	[1.89]	[1.90]	[1.61]	
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EPI Quintiles					EPI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	341	342	342	340	334	26	29	33	25	24	
	(2)	97	95	96	96	94	50	49	51	50	48	
	(3)	69	69	69	69	69	86	85	87	88	89	
	(4)	60	60	60	60	60	186	189	199	196	200	
(5)	56	56	56	56	55	1308	1408	1476	1529	1779		

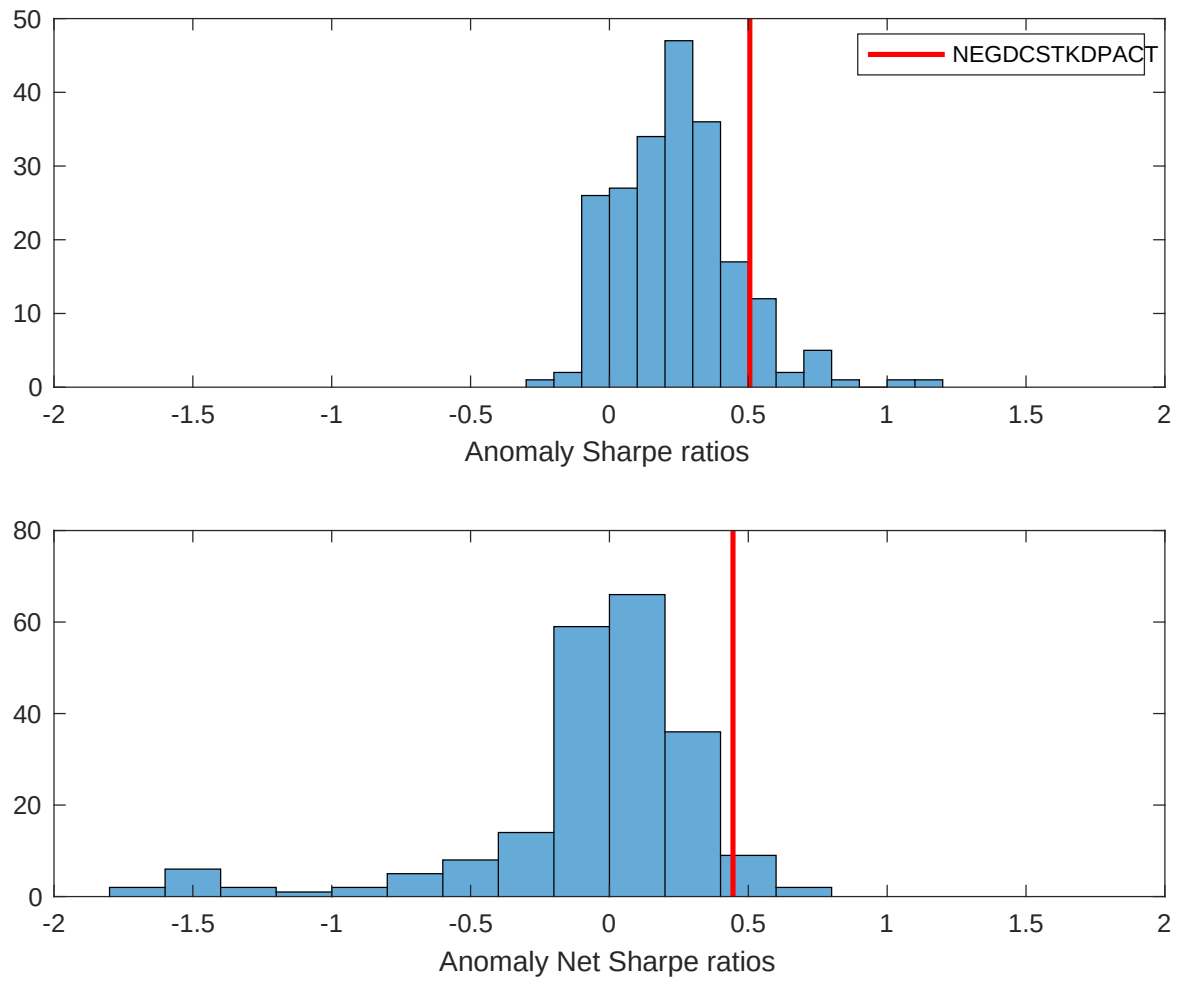


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EPI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

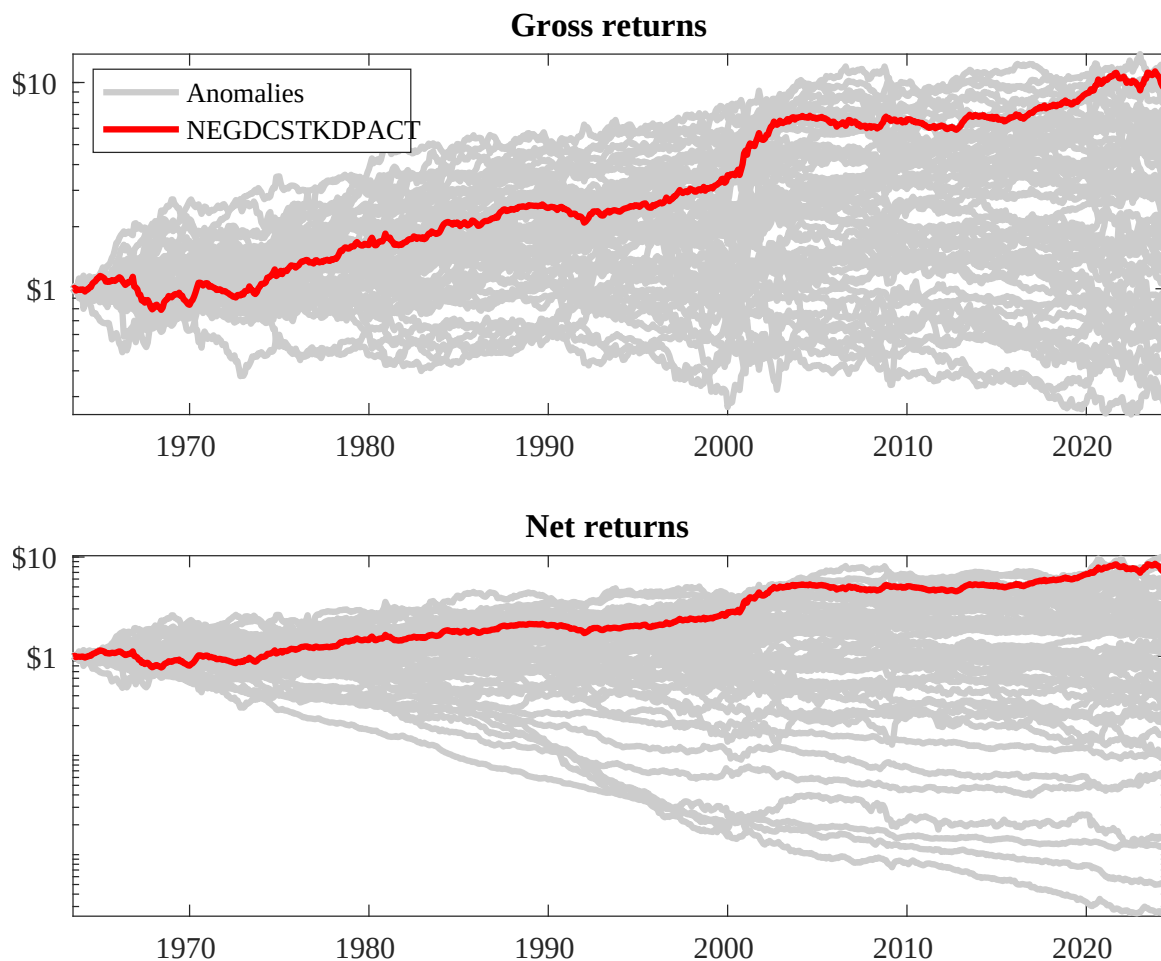
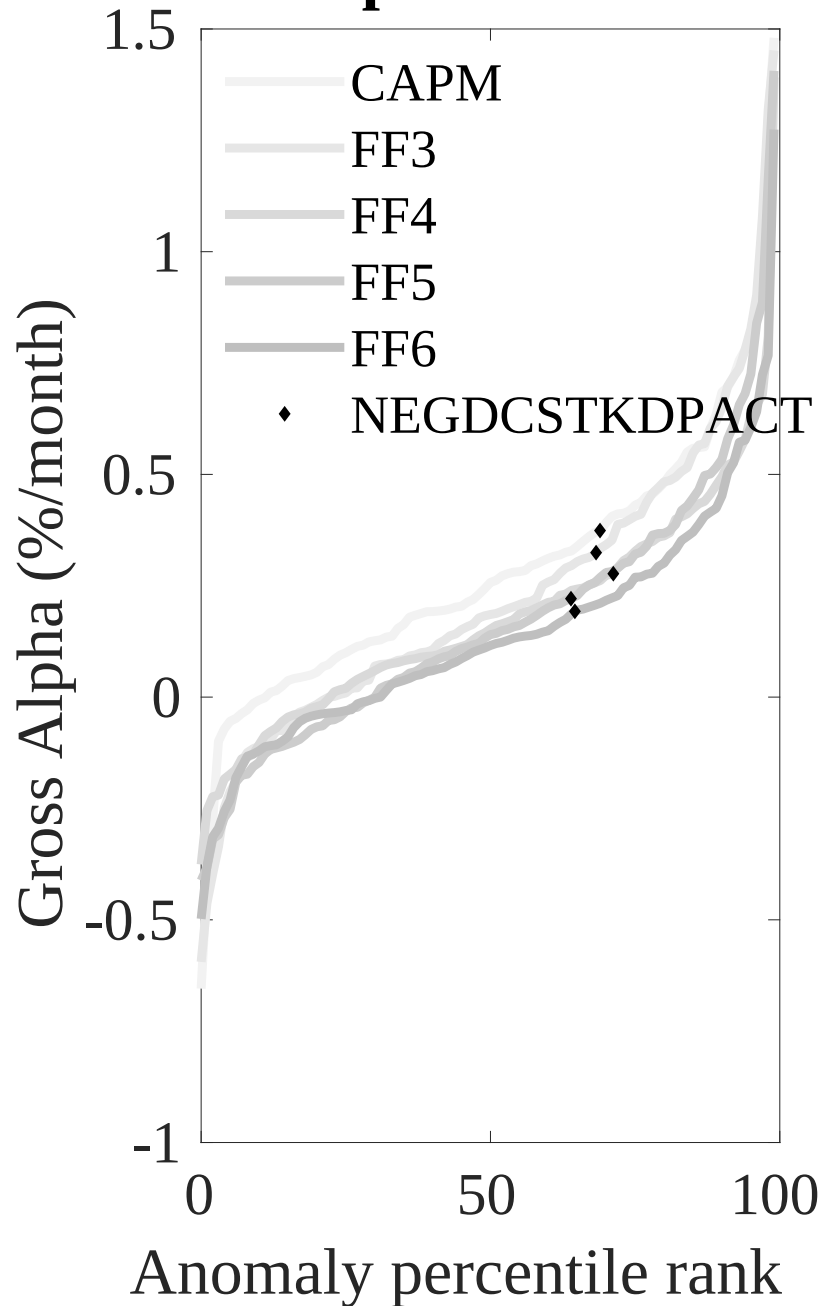


Figure 3: Dollar invested.

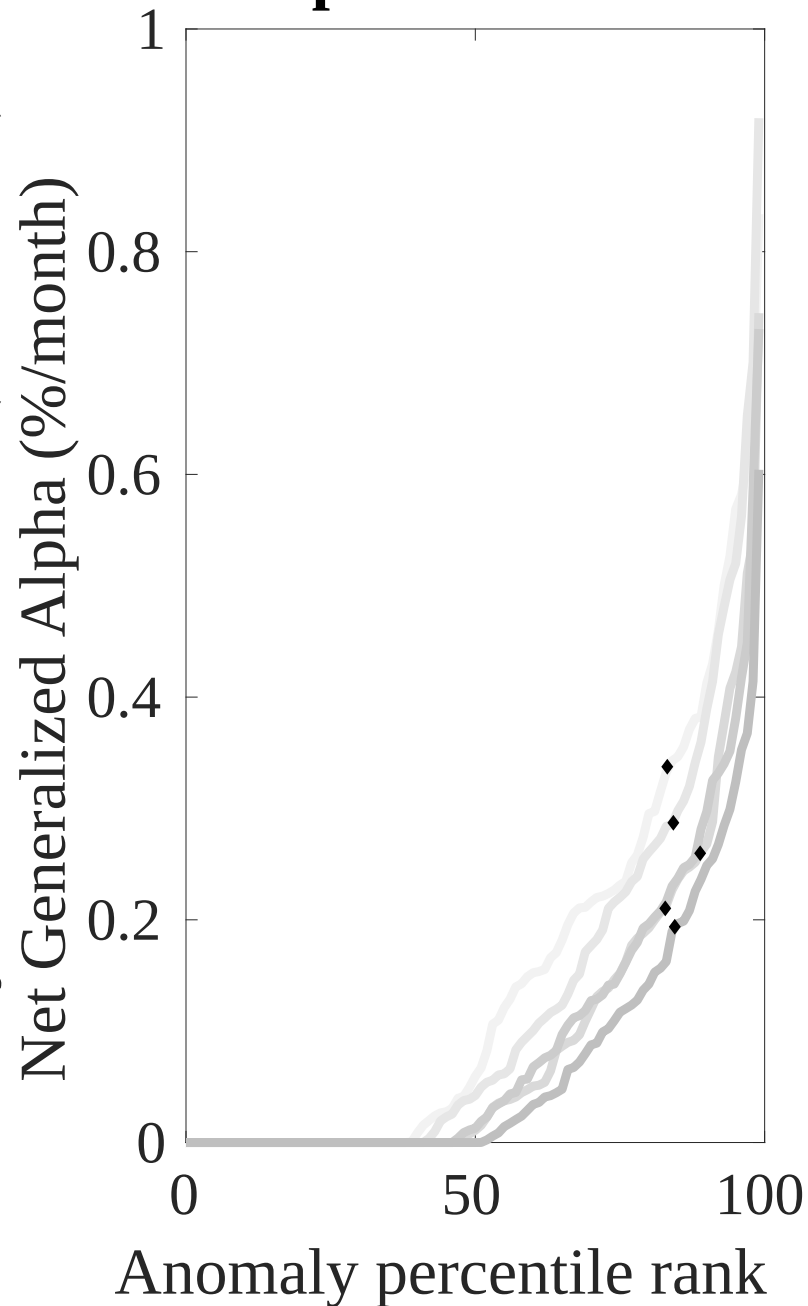
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EPI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



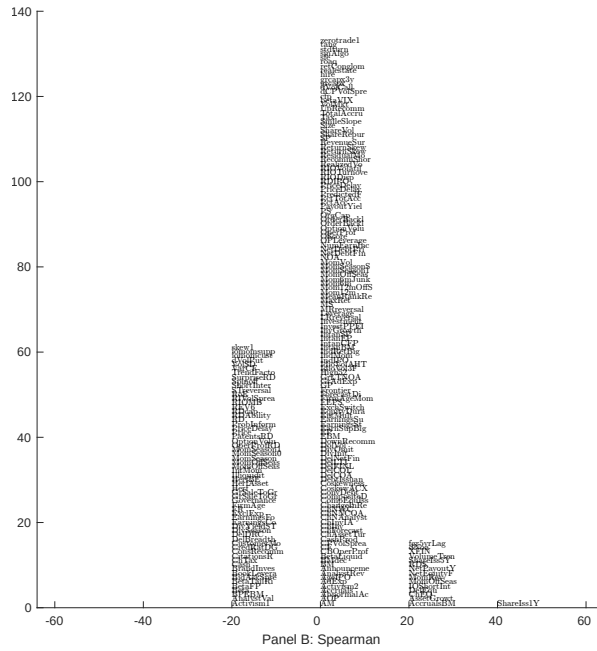
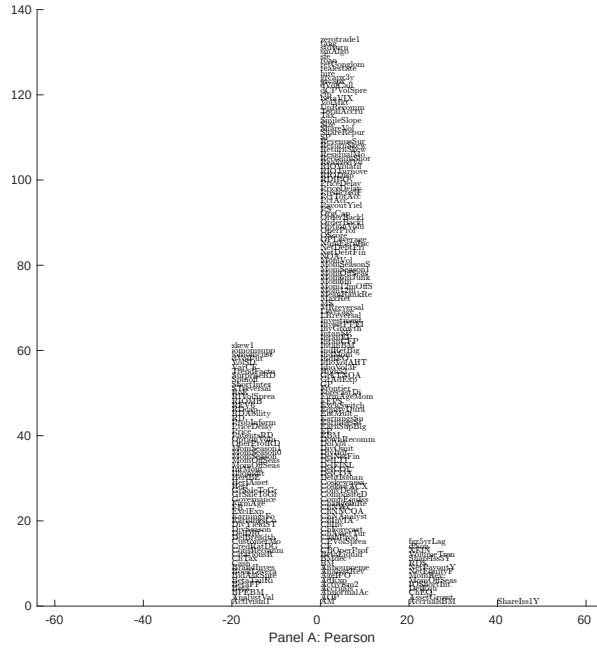


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EPI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

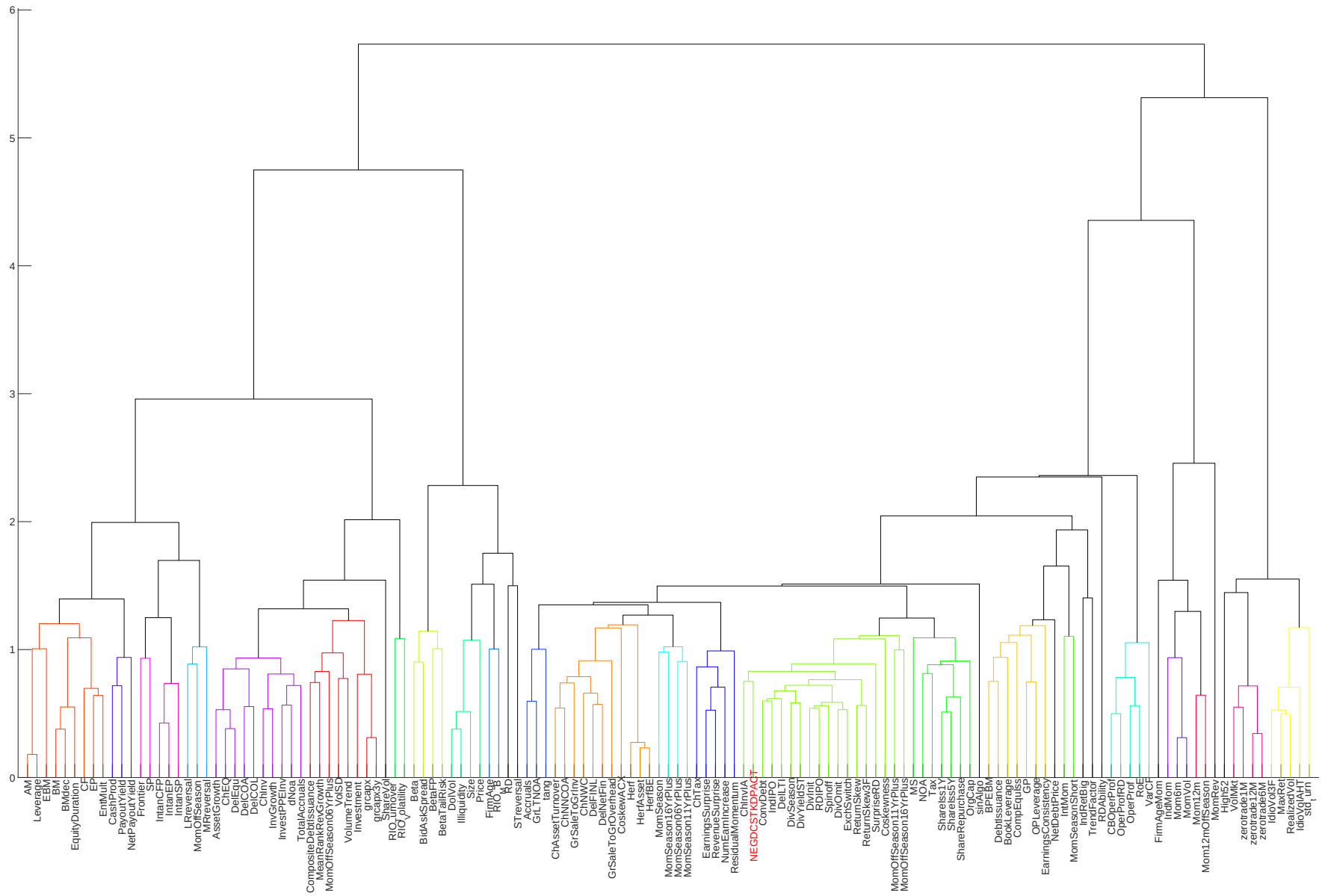


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

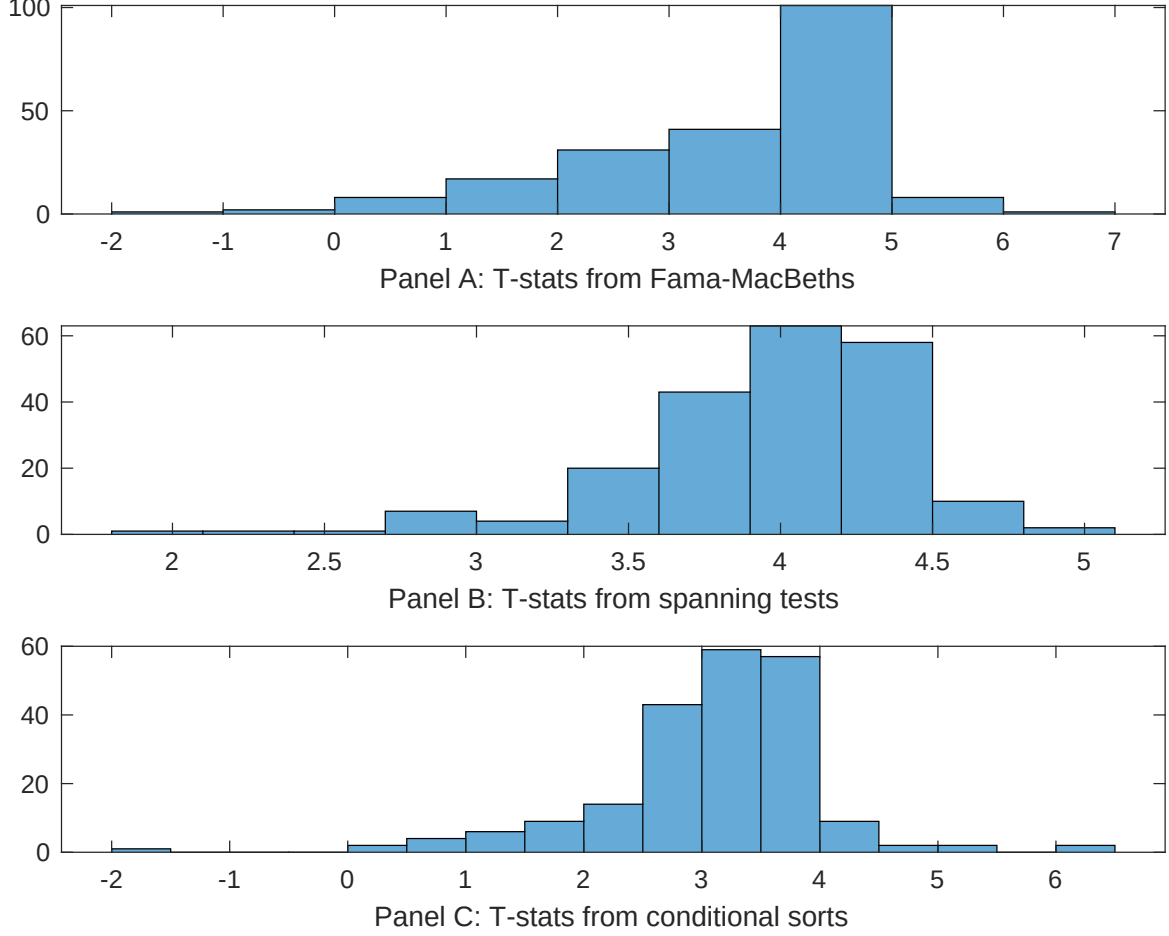


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EPI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EPI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EPI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EPI. Stocks are finally grouped into five EPI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EPI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EPI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EPI}EPI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.98]	0.12 [5.43]	0.17 [6.64]	0.13 [6.37]	0.13 [5.61]	0.13 [5.42]	0.14 [5.00]
EPI	0.74 [3.87]	0.45 [2.07]	0.71 [3.32]	0.69 [3.66]	0.81 [3.92]	0.75 [3.47]	0.44 [1.90]
Anomaly 1	0.14 [6.01]						0.85 [3.78]
Anomaly 2		0.21 [1.88]					0.12 [2.60]
Anomaly 3			0.38 [3.21]				-0.25 [-0.16]
Anomaly 4				0.27 [4.73]			0.27 [5.88]
Anomaly 5					0.12 [3.17]		0.98 [1.63]
Anomaly 6						0.14 [2.45]	-0.15 [-1.85]
# months	715	715	720	715	720	630	623
$\bar{R}^2(\%)$	0	1	0	1	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EPI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EPI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18 [2.33]	0.23 [2.97]	0.18 [2.35]	0.12 [1.63]	0.19 [2.48]	0.23 [2.84]	0.19 [2.43]
Anomaly 1	30.92 [8.10]						13.48 [2.64]
Anomaly 2		19.34 [6.52]					12.96 [3.32]
Anomaly 3			33.34 [8.18]				38.69 [6.22]
Anomaly 4				27.98 [6.86]			14.80 [2.93]
Anomaly 5					13.83 [3.57]		-18.62 [-3.26]
Anomaly 6						7.35 [1.96]	-17.37 [-3.79]
mkt	-0.07 [-0.04]	2.16 [1.15]	0.29 [0.16]	0.43 [0.24]	-1.06 [-0.57]	-0.41 [-0.20]	0.05 [0.02]
smb	-0.17 [-0.07]	1.75 [0.65]	-2.46 [-0.95]	0.45 [0.17]	-1.52 [-0.57]	5.96 [1.84]	-1.23 [-0.40]
hml	-1.24 [-0.36]	-7.95 [-2.13]	-4.43 [-1.26]	1.87 [0.53]	-2.15 [-0.59]	0.03 [0.01]	-3.81 [-1.03]
rmw	-9.97 [-2.45]	-4.49 [-1.13]	8.35 [2.38]	-3.24 [-0.84]	8.04 [2.21]	9.27 [2.10]	2.28 [0.49]
cma	19.82 [3.54]	22.74 [3.98]	6.73 [1.04]	25.41 [4.61]	25.12 [3.83]	22.65 [3.76]	-6.08 [-0.88]
umd	2.81 [1.57]	5.76 [3.17]	3.65 [2.05]	2.40 [1.32]	4.59 [2.49]	5.15 [2.70]	3.79 [2.09]
# months	728	728	732	728	732	630	626
$\bar{R}^2(\%)$	22	20	22	20	16	12	24

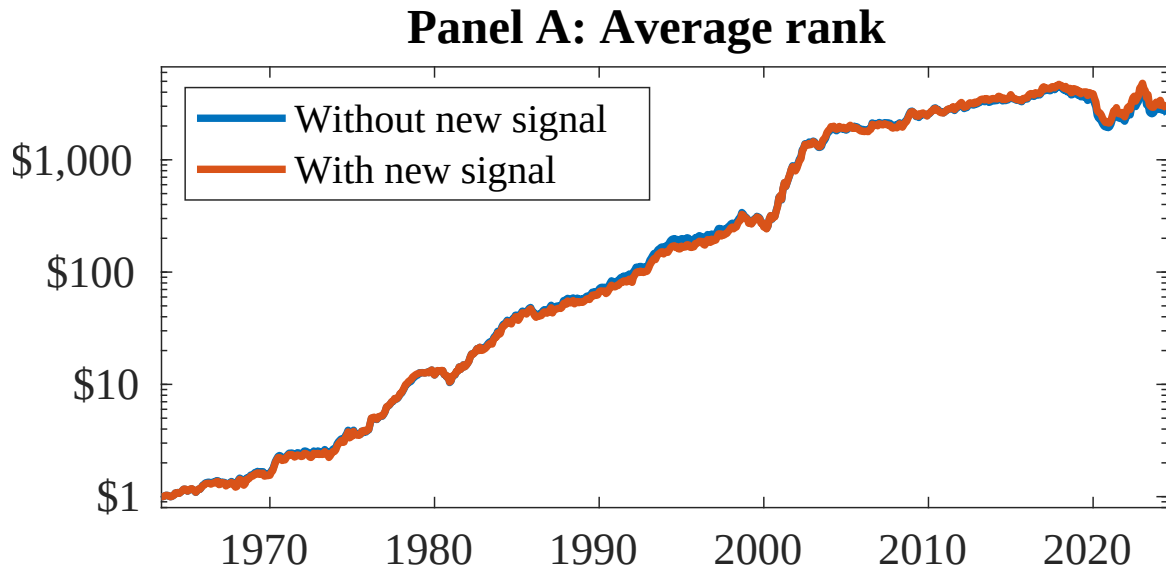


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EPI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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